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Early Detection of Anemia Using Ensemble Machine Learning Algorithms with Data Balancing

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***Abstract*— A global public health concern that impacts not just the impoverished but also rich populated countries is anemia. It's an international concern for public health assurance. Although its primary reason is iron deficiency but also other causes, such as major malnutrition, insomnia, stress, etc. This study intends to predict anemia more effectively so that early diagnosis can save lives. The dataset was collected from Kaggle and there were 8544 data. SMOTE was preprocess the dataset in order to address the imbalance and improve model performance and MinMaxScaller to normalize and Standard Scalar to standardize the dataset so that Machine learning (ML) algorithms performs better. In this proposed technique, we have used five ML algorithms: SVM, KNN, random forest classifier, logistic regression, naive Bayes, and voting classifier for ensemble. Among these techniques, we combined three distinct algorithms: random forest, logistic regression, and SVM classifiers using an ensemble learning approach with soft voting. The result becomes more accurate. With precision, recall, and FI score and 99.94% accuracy rate. Although there are machine learning methods for detecting anemia, they frequently overlook three important issues: 1) addressing class imbalance in clinical datasets, 2) preserving robustness across a range of patient demographics, and 3) offering clinically useful predictions that are easy to understand. In order to overcome these constraints, this study introduces a revolutionary soft voting ensemble that integrates SVM, RF, and LR in a unique way that will fulfill all the major advancements.**

Keywords*— *Anemia, Classification Algorithms, Voting classifier, Prediction, Ensemble, Machine Learning.

I. INTRODUCTION

Anemia is a global health issue. Without enough iron, the body cannot make enough hemoglobin for the human body which mainly creates red blood cells. It mostly affects women and children. Anemia occurs when the organ and tissues don't get enough oxygen. Anemia can be caused by many reasons like malnutrition, malaria, organ failure, accident, liver disease etc. Anemia is preventable and treatable if it is cured at an early stage. Anemia comes from the ancient Greek term anime, which means "lack of blood" [1]. Anemia is the medical term for a reduction in the quantity of red blood cells in circulation. It is crucial to have a medical professional identify the precise cause of anemia. It can be caused by various conditions. Among the possible reasons are Malnutrition, Drugs, Periods or heavy menstruation, being pregnant, gastrointestinal tract bleeding issues with the production all of the bone-marrow's blood cells, including red blood cells [2]. Complications unique to red blood cells such as hemoglobin-related problems (such sickle cell disease and thalassemia). Symptoms of anemia include exhaustion, a decreased ability to do strenuous labor, and dyspnea. Poor nutrition and other health issues are indicated by anemia. The symptoms are fatigue, lightheadedness or dizziness, as well as cold hand and feet headache and dyspnea particularly after exerting oneself. More severe signs of severe anemia include mucosal membranes that are pale (in mouth, nose, etc.) pale complexion, quick breathing, elevated heart rate, lightheadedness when standing, and easier bruising under the fingernails [3].

Worldwide influence: the effects of anemia can differ. Through behavioral disturbances such reduced motor activity, social engagement, and attenuation to activities as well as development delays, it can affect adult productivity, overall quality of life and academic success. Maternal mortality, poor birth weight, and premature birth are some of the adverse effects of anemia during pregnancy. In addition to its detrimental consequences on health, anemia may have a substantial financial impact on individuals, families, communities, and countries. Three million mothers and babies died in 2013, according to the World Health Organization (WHO). 40% of expectant mothers and 42% of children under six are anemic globally, according to a WHO study [4]. Mostly a test know as a complete blood count, or CBC, is used to diagnose anemia. In particular, physicians use the CBC's hemoglobin or hematocrit findings to identify anemia and monitor how well a patient is responding to

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therapy. This illness, which affects 33% of individuals globally, is caused by iron deficiency.

The use of machine learning algorithms is one method for the non-invasive diagnosis or identification of clinical disorders nowadays. Every day, enormous volumes of data are produced by contemporary medical systems. To uncover hidden patterns and glean valuable information, this data must be mined and examined. Data mining is the process of identifying novel patterns in data collected from various sources. Numerous ML methods are employed in predictions of healthcare. An important element of medical study is the prediction of various illnesses and the factors that cause them. Health data is utilized in the medical industry to identify diseases, forecast epidemics, enhance quality of life, and stop premature mortality [5]. Three distinct categorization techniques for prediction will be examined in this work. While classical statistical methods try to identify relationships between variables, algorithms concentrate on generating predictions as accurately as possible. Machine learning algorithms are data-driven and devoid of presumptions, in contrast to conventional statistical methods. In the healthcare industry, machine learning outperforms conventional techniques in disease prediction [6]. The technique known as ML allows machine to learn from experience and data in order to make judgments and predictions [7]. Supervised learning, classification algorithms using an ensemble approach and hybrid models will be used in systems scope to improve accuracy and make a trustworthy anemia prediction. In order to provide an anemia prediction model with an optimal generalized performance of greater than 99.94% accuracy, the goal is to balance bias and variance. Since there is already a web tool for anemia prognosis, our study assists with a broad user base understanding and diagnoses using fundamental clinical records more accurately and efficiently. It takes a lot of resources and is difficult to diagnose anemia, especially in rural areas where resources are limited [8].

Improved machine learning performance in forecasting anemia has been demonstrated in earlier research [9]. In this proposed method we ensemble algorithms with voting classifier to detect anemia more accurately. The dataset was preprocessed with Smote, MinMaxScalar, and Standard Scalar to make the accuracy better. Thus, the purpose of this research is to diagnose anemia early so that we can decrease the risk factor in the population. The result of this analysis, in our opinion, will make it simpler for medical professionals to recognize, treat, and refer to those who have anemia. Although there are still significant limitations, recent research has shown that machine learning has the capacity to diagnose anemia [10–15]. First, the majority of methods rely on a single model that is susceptible to certain data biases. Secondly, it is uncommon to take into account the trade-offs between model complexity and real-world deployability in environments with limited resources. The field is advanced by our efforts by the robust ensemble model.

II. Literature Review

Yeruva et.al. [10] Talk about the application of deep neural networks for sickle cell anemia early diagnosis, emphasizing the significance of early detection to avoid severe discomfort, organ damage, and a shortened life expectancy. To precisely detect sickle cells, data mining techniques are recommended, with an emphasis on the multilayer perceptron (MLP) classification algorithm. The study uses a dataset from the Thalassemia and Sickle Cell Society (TSCS) in India to compare the MLP classification method with other well-known mining and machine learning algorithms. Additionally, a thorough analysis of sickle cell crises, a variety of computational techniques, and the development of a web application using TSCS data are provided. The authors stress the drawbacks of manual diagnosis and evaluation and propose the use of MLP classification algorithms for accurate detection. In this methodology the accuracy was 99%. It's a time consuming approach and it's prone to overfitting.

Sasika et.al. [11] Delve into the significance of using data mining methods for forecasting sickness anemia, emphasizing the potential uses of machine learning methods in the medical domain. It provides a comprehensive overview of the approach. Performance metrics, findings and conclusions, eventually highlighting the Random Forest tree's efficacy technique for forecasting anemia from the data and its potential for wider medical use. Anemia is a worldwide health issue that has social and economic repercussions. Using the scrum technique, a web system was created to identify anemia in youngsters from Peru. The system employs picture analysis algorithms and makes food recommendations. Scrum incorporates a number of processes and integrates roles across teams. Their accuracy was 97.57% To identify the existence of anemia, which is defined by low hemoglobin levels, Jayakody [12] went over a non-invasive technique for measuring the concentration of hemoglobin in human blood. Hemoglobin tests typically use needles to draw blood samples, but the suggested method makes use of an algorithm based on neural networks that were created using pictures of patient's fingertips that were shot with a webcam. This algorithm is used by a noninvasive device to show hemoglobin levels. Patients fill out a questionnaire and register for an account on a mobile app prior to the surgery. The outcomes from early -stage anemia detection is made possible by processing the device and the mobile app using a machine learning technique to produce a result. The accuracy of their result was 95%.

A prediction model for iron deficient anemia and moderate to severe anemia in patients with recently diagnosed ulcerative colitis was created in Dixit et.al. Proposed method [13]. Multivariable logistic regression was employed to forecast the prevalence of these illnesses using a national dataset of 789 patients. Age, albumin level, and mild anemia at UC diagnosis were well included in the model, which had an AUC of 0.69. Age, albumin level, mild anemia, and African American ancestry were predictors of moderate to severe anemia with iron deficiency. The model's performance was validated internally, and its calibration was consistently good.

The authors of [14] want to use the Complete Blood Count (CBC) to predict anemia in children under five by creating a model with four machine learning algorithms. Machine learning is a novel method for disease prediction. The best-performing classification model was Random Forest, out of the six that were chosen for testing. To increase accuracy, boosting techniques and stacking ensemble learning methods were applied. It has been demonstrated that

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bagging is an effective method for improving accuracy for the top performance in both balanced and unbalanced data. To increase the accuracy of predicting anemia in children under five, a new prediction framework was devised. Their estimated accuracy is 99.22%.

Using machine learning techniques, [15] created predictive models for children's anemia; Random Forest demonstrated higher accuracy. Age, residential location, gender, maternal age, income, education, employment position, and the mother's health traits are all factors linked to anemia. The best predicting models were random forest and logistic regression. The goal of the suggested approach is to create a quick and accurate technique for identifying and measuring anemia utilizing a system design that incorporates user registration. Data entry, evaluation. Their accuracy is 68.75%. Table 1 shows the methodology used and its analysis in the context of anemia diseases over various background works.

III. Proposed Methodology

In order to increase accuracy and make the anemia prediction reliable, the suggested method employs hybrid models or an ensemble approach shown in "fig. 1". We build the anemia prediction model so that medical professionals can detect anemia as early as possible. Supervised learning is a type of machine learning that teaches computers to predict outputs by providing them with appropriately labeled training data. Some input data has already been given the proper output since the data is labeled, normalized, scalarized. The training data that machines are given serve as the supervisor in supervised learning, which teaches them to correctly predict an output. It makes use of the same concept as when a student is instructed by a mentor. The act of providing the ML model with precise input and output data is known as supervised learning. The goal of a supervised learning method is to identify a mapping function that connects the input variable (X) and output variable (Y). The proposed architecture to detect anemia is shown in "Fig. 1".

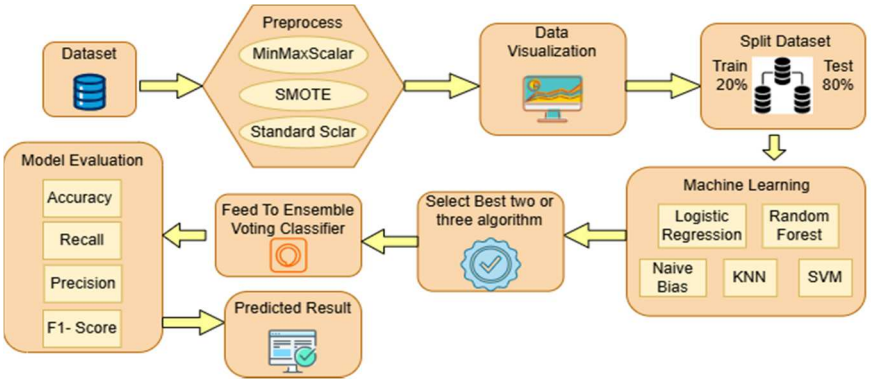


Fig 1: Anemia Prediction Model

A. Dataset

Gathering patient information from several sources is a component of data extraction. To find trends or insights, the analysis entails processing and analyzing the data that has been retrieved. The dataset used for this methodology through all this examination was collected from a public web portal Kaggle and this dataset is especially for predicting anemia, this contents 8544 data and it has five attributes and result: Gender, HGB, MCV, MCH, and MCHC [21]. The correlation between the attributes are shown in "fig 2". Hemoglobin, often known as Hgb, is an iron-containing protein that helps red blood cells carry oxygen. Red blood cells get their characteristics red hue from hemoglobin. An animal's metabolism is powered by aerobic respiration, which is made possible by hemoglobin in the blood, which transports oxygen from the respiratory organs from the respiratory organs (lungs or gills) to the various tissues of the body. For every 100 milliliters of blood, a healthy human contains 12 to 20 grams of hemoglobin. A chromoprotien, a metalloprotien and a globulin are all componenets of hemoglobin. Hemoglobin is a metalloprotein, a chromoprotein, and globulin. The average amount of hemoglobin found in one red blood cell is known as the MCH value.

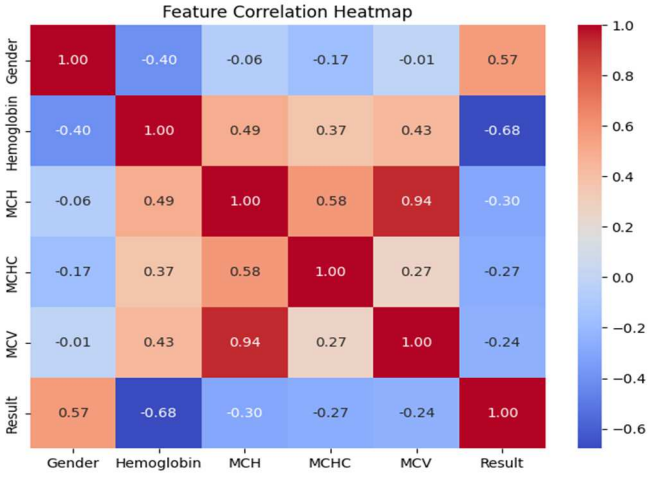


Fig 2: Feature Correlation Heatmap

The average amount of hemoglobin inside a specific volume of red blood cells is known as the mean corpuscular hemoglobin concentration, or MCH. Liver illness, hyperactive thyroid, and some forms of anemia are among the

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causes of elevated MCHC. MCV is a blood test measures the average size of your red blood cells. An MCV blood test is a part of complete blood count (CBC), a group of tests that measures the number and type of cells in your blood. It is used for general blood checks. And in dataset there is gender and outcome column.

B. Data Preprocessing:

Preparing raw data for analysis by cleaning, converting, and preparing it is known as data - preprocessing. It involves activities including feature scaling, normalization, addressing missing values**,** and categorical variable encoding. SMOTE, MinMaxScaller, and Standard Scalar are employed in this suggested model as preprocessing techniques. Normalization and Standard Scalar help with removing biasness and handles outlier better and works well for ML models. SMOTE helps to balance the imbalance dataset and prevent overfitting by removing duplication of data. It is a crucial and important stage in data analysis that improves the model's dependability and performance.

C. Machine Learning Models

The bias and variance will be taken into consideration while choosing the model. The data has been divided up for training 20% and the model has been trained for will be supplied into algorithm. We have used 5 ML classifiers: Random forest, SVM, K-Neighbor Classifier, Naive Bayes classifier, Logistic Regression to predict anemia. Here we combine the algorithms into an ensemble method to achieve balanced bias, variance and higher accuracy. An ensemble model is a type of machine learning technique that aggregates the prediction of multiple distinct models to provide predictions that are more reliable and accurate than any single model alone. Using the diversity and combined intelligence of multiple models, ensemble modeling aims to improve prediction performance. Ensemble models are used in many machine learning techniques and application. Task such as: grouping, regression, and classification. Ensemble models include several features, such as: increasing accuracy, robustness, decreased variance and bias, and flexibility. The parameter used in these algorithms have been given in Table-I.

TABLE I. HYPERPARAMETER VALUES FOR EACH CLASSIFIER

Classifier	Parameter	Value
Support Vector Machine (SVM)	gamma	Scale
	C	1.0
	Degree	3
Random Forest	n_estimators	42
	max_depth	None
KNN	n_neighbors	5
Logistic Regression	C	1.0
Gaussian Naive Bayes	No parameter grid needed for Naive Bayes	
Voting Classifier	Weights	Soft

As soon as the anemia will be predict the patient can take all possible step and medication and other facilities through taking care to cure.

The ensemble method is a method that work with combining two or three model to get better result or value. It combine the prediction of multiple models and the weakness of single model removed. It make the ML algorithm work better than single ML algorithm. Ensemble with voting makes the system result better than any other approach. Soft voting reduce the bias and variance and improve the robustness. A single method gives poor result and it does not give accurate and precise value. The RF, SVM and LR model with soft voting works very well and give accurate and precise value than others ML. The LR model's goodness-of-fit can be evaluated in number of ways. RF is a simple and widely used algorithm. SVM is a powerful and versatile algorithm that widely used for high effectiveness and remove overfitting. Using these model with voting give a powerful result to detect anemia.

D. Environmntal setup

After being collected, the data was divided into training and testing sets and then jumbled at random. To avoid overfitting and data leakage, specific preparation techniques were applied prior to the testing sets being employed. To get the desired results, the study needed a certain setup, and Table-II lists the setup requirements as well as the requirements for conducting the research.

TABLE II. SYSTEM'S SPECIFICATIONS

Platform	Google Co-Lab
Language	Python
Libraries	Matplotlib, Scikit-Learn, NumPy, Pandas, and Seaborn
Configuration	
RAM	16 GB
CPU	Core i7, 12th Gen
Session Limit	12 h
Disk Space	512 GB

E. Performane Metrics

Accuracy, precision, recall, and F1-score were among the measures used to assess how well the categorization techniques performed when applied to the anemia data. The terms T, F, P, and N stand for true, false, positive, and negative outcomes, respectively, in the confusion matrix analysis. For example, TP represents the total number of correctly classified healthy points of data. To calculate the overall accuracy formula (1) is used. The positive prediction can be calculated using formula (2). F1- Score metric, which assesses the harmonic mean and precision. Last equation gives the mathematical definition of recall [21].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$F1 - Score = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3)$$

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$$Recall = \frac{TP}{TP + FN} \quad (4)$$

IV. Result Analysis

Confirmed findings were obtained from testing the suggested anemia prediction and estimate models in a variety of settings. A Random Forest classifier, SVM (Support Vector Machine), KNN (K-Nearest-Neighbor), Naive Bayes classifier, Logistic Regression, Decision Tree and Voting Classifier are the six models used in this methodology. The accuracy metrics of the ML models used on the training data are shown in Table-III.

TABLE III. Evaluate Different Technique of ML

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Support Vector Machine	99.18	99.18	99.18	99.18
K-Nearest Neighbors	99.00	99.00	99.01	99.00
Logistic Regression	97.01	97.01	97.02	97.01
Naive Bayes	91.69	89.40	85.71	87.52
Voting Classifier	99.94	99.94	99.94	99.94

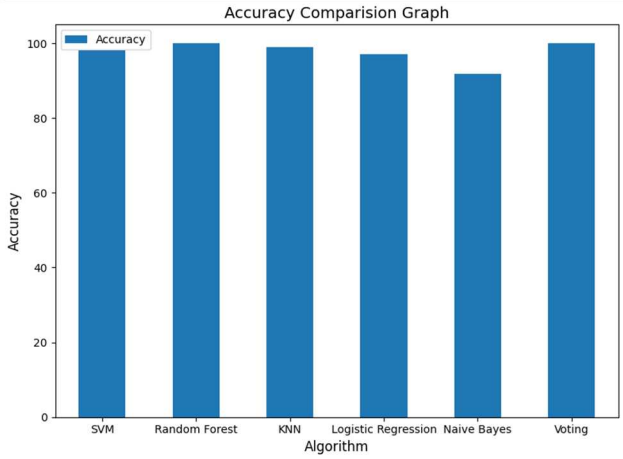


Fig: 3 Accuracy Comparison Graph

Since a portion of the models in fig-3 above exhibit overfitting, we will adopt the new technique of employing an ensemble voting classifiers to lessen overfitting. We will select the top three algorithms from the list of algorithms above that are relevant to the research topic. The chosen three algorithms are: For both linear and nonlinear classification, SVM is potent Ml technique that is often utilized. Because SVM concentrates on determining the largest separation hyperplane between many classes, it is especially successful. One of the most widely used and straightforward classification and regression classifiers in ML nowadays is RF. Logistic Regression is widely used for binary classification tasks. As in our dataset we have binary value, it worked well, hence these three methods were selected. When mixing these three techniques lessen the model's overfitting and using hard voting make more precise and accurate value. The performance of the ensemble model is shown in below.

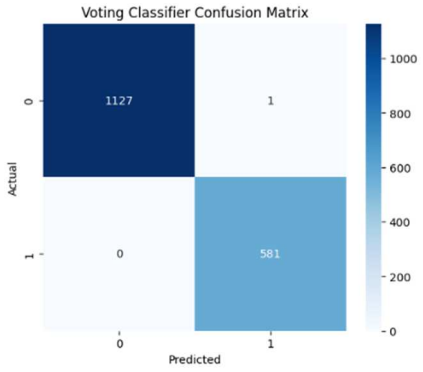


Fig 4: Confusion Metrics of SVM, RF, LR with voting.

The above matrix shows the accuracy and the true positive, true negative, false positive, false negative vale of the 3 algorithms with soft voting ensemble method. The ROC curve has been given below to show the comparison between the models with Logistic Regression most favorable curve in fig 4.

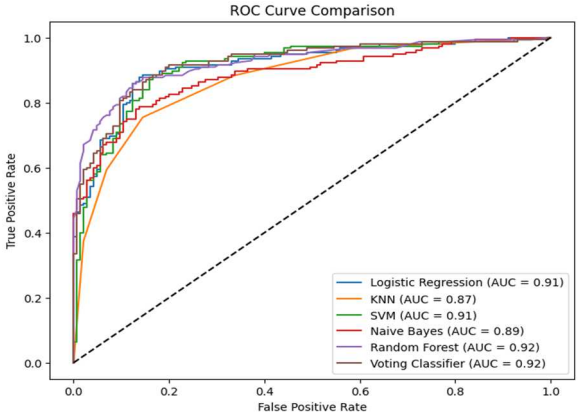


Fig. 5. ROC curves of different ML classifiers

Ultimately the result is shown that Logistic Regression and Random Forest and SVM have done the most favorable and precise result 99.94%. In this 5 algorithms the Logistic Regression, Random Forest and SVM has done more accurate value. The Logistic Regression has 97.01% and SVM has 98.31% accuracy and Random forest has 100%. So all these three with soft voting make an accurate and precise value. The ultimate result show with SVM and Logistic Regression and Random Forest with soft voting 99.94% value. Table IV evaluate the results of multiple ensemble models. The average accuracy is up to 95% and 99%.

TABLE IV: Evaluating Multiple Ensemble Model

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM, RF,LR	99.94	99.94	99.94	99.94
Log_Reg, Naive Bias	97.60	97.59	97.60	97.59
SVM, KNN	99.12	99.12	99.12	99.12
RF, SVM	99.88	99.88	99.98	99.98
KNN, Log_Reg	99.24	99.24	99.24	99.24

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Table-V is the overview of several ML methods that determines the accuracy of the models in several studies. The proposed model gained the highest accuracy among them.

TABLE V: EVALUATING SEVERAL TYPES OF MACHINE LEARNING METHODS TO DETERMINE WHETHER ANEMIA EXISTS

Authors	Data modality	Dataset used	No. of Instances	Classification	Accuracy
[10]	Clinical data	TSCS-	-	MLP	99%
[11]	Clinical data	-	-	Random Forest	97.57%
[12]	Clinical data	-	-	Counting hemoglobin with Ml	95%
[23]	Clinical data	Kaggle online website	1422	Ada Boost, Random Forest	98.7%
Proposed Model	**Clinical data**	**Kaggle online**	**8544**	**Ensemble** (SVM, RF, LR)	**99.94%**

V. CONCLUSION

Accurate and early prediction of anemia is crucial for effective disease management and improved patient outcomes. In this study, we leveraged ensemble machine learning technique to develop a robust predictive model. The proposed prediction method is effective approach for early anemia prediction using ensemble machine learning. This study has demonstrated high accuracy 99.94% and reliability. Showing the potential of machine learning in developing healthcare diagnostics. By preprocessing we ensure a robust and reliable dataset and the algorithms evaluated RF, SVM, LR with ensemble model which delivered perfect accuracy 99.94%.The outstanding results of our machine learning model, which have been painstakingly developed and carefully verified using actual real data, and large in numbers amply illustrate the effectiveness of our approach. In order to further increase the models transparency and reliability. The investigation believes that Explainable AI (XAI) is a crucial improvement. Better understanding of feature significance and model choices can be obtained by using XAI approaches like SHAP and LIME. Better clinical decision making will result from medical workers being able to comprehend and accept the AI's forecast. These explainable techniques will be investigated in future studies to guarantee the models dependability and moral application in medical diagnostics.

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